



Y







► Given a particle of mass  $m \geq 0$  with position  $q \in \mathbb{R}^3$  and momentum  $p = mv \in \mathbb{R}^3$

$$H(x) = m g q + \frac{1}{2m} |p|^2 = \frac{m}{2} |v|^2 + C =: U(v)$$

▶ The potential energy is constant, so the energy of the system is just the sum of kinetic energy

► Here, the reference corresponds to still particles with  $\eta(dx) \equiv dx$



$$\pi_{\beta}(v)=\left(\frac{\beta m}{2\pi}\right)^{\frac{3}{2}}\exp\left(-\frac{\beta m}{2}|v|^2\right)$$

$$\pi_{\beta}(s) = 4\pi \left( \frac{\beta m}{2\pi} \right)^{\frac{3}{2}} s^2 \exp \left( -\frac{\beta m s^2}{2} \right)$$

X



$$\beta - 1 = k_B T$$



*P*

—

*nmv*

► For a particle with speed  $s \equiv |v|^{1/2} \geq 0$  we have

$$\propto s^2 \exp\left(-\frac{ms^2}{2k_B T}\right)$$



# EXAMPLE: GAS IN A BOX

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- ▶ Given a particle of mass  $m > 0$  with position  $q \in \mathbb{R}^3$  and momentum  $p = mv \in \mathbb{R}^3$
- ▶ The potential energy is constant, so the energy of the system is just the sum of kinetic energy

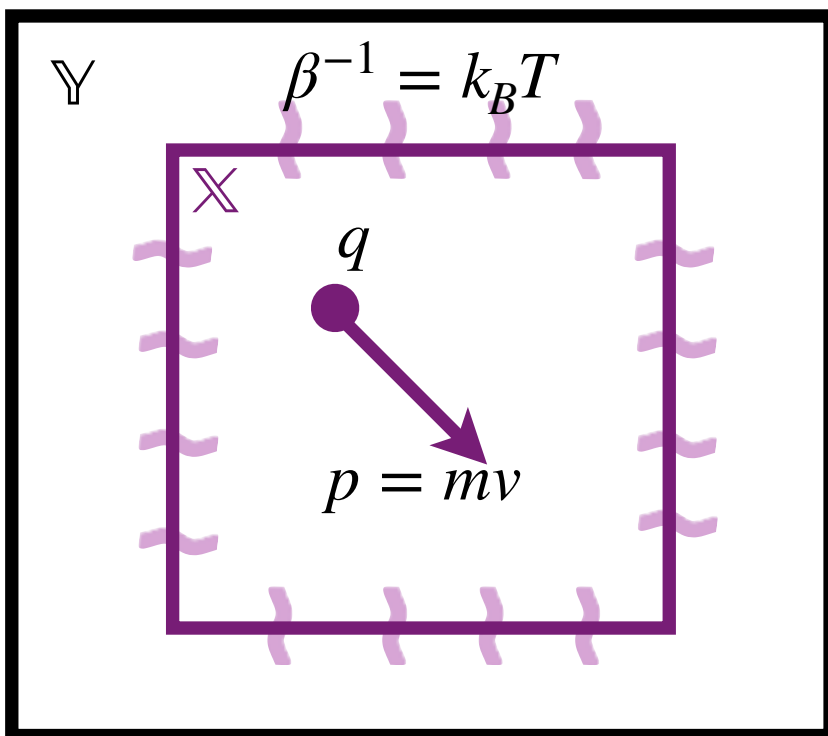
$$H(x) = mgq + \frac{1}{2m} |p|^2 = \frac{m}{2} |v|^2 + C =: U(v)$$

- ▶ Here, the reference corresponds to still particles with  $\eta(dx) = dx$

$$\pi_\beta(v) = \left( \frac{\beta m}{2\pi} \right)^{\frac{3}{2}} \exp \left( -\frac{\beta m}{2} |v|^2 \right)$$

- ▶ For a particle with speed  $s = |v|^{1/2} \geq 0$  we have

$$\begin{aligned} \pi_\beta(s) &= 4\pi \left( \frac{\beta m}{2\pi} \right)^{\frac{3}{2}} s^2 \exp \left( -\frac{\beta m s^2}{2} \right) \\ &\propto s^2 \exp \left( -\frac{m s^2}{2k_B T} \right) \end{aligned}$$



# EXAMPLE: GAS IN A BOX

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$$\pi_{\beta}(s) \propto s^2 \exp\left(-\frac{ms^2}{2k_B T}\right)$$

